

EERA: Externalizing the Eye to support embodied Reflection on eye care Awareness

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Figure 1: (a) EERA in use. (b) The externalized eye displayed as a wearable, shareable object. (c) Two participants holding and observing each other's eye representations. (d) A pair of participants engaging in a card-based social interaction using only eye representations as visual cues.

Abstract

We present EERA, an interactive device that externalizes the user's eye as a shareable object. Through real-time magnified visualization of eye movements, autonomous blinking, and the exchange of eye representations between participants, EERA enables people to observe, feel, and socially engage with their own eyes in unusual ways. The device was developed through a participatory design workshop involving industry research organizations and students from a school for the Deaf and Hard-of-Hearing, where visual information plays a central role in daily communication, motivating a visually salient and embodied interaction design. We conducted

a user study and collected post-experience questionnaires after a short sequence of playful tasks such as emotion guessing games that designed to externalize the eye, disrupt bodily agency, and introduce self-other comparison. Through the experience and feedbacks we received, we explore how externalizing a normally unseen body part can support embodied reflection and foster new perspectives on everyday eye care during daily social interaction.

CCS Concepts

• **Human-centered computing** → **Interactive systems and tools**.

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1 Introduction

In recent years, the World Health Organization's report shows that eye health is a significantly neglected area of public health, even though 2.2 billion people have a near or distance vision impairment [11]. They also stated that at least 1 billion out of the vision impairments cases could have been prevented. Although visual cues, including facial expressions and gestures, are estimated to account for up to 55% of the impact in interpersonal communication, underscoring the critical role of vision in daily life, eye health remains frequently neglected. In addition, the annual global productivity losses associated with unaddressed vision impairment are estimated to be US \$410.7 billion.

We believe the lack of awareness about eye health and eye care is one of the possible reasons for this gap. Eye health is often treated as an abstract or informational issue rather than a bodily and experiential one. As a result, conventional approaches to promoting eye care, which primarily rely on information provision or recommendations, may fail to foster a sense of personal relevance or embodied understanding.

We look to HCI interactive system for potential solutions: HCI has seen an increasing interest in interfaces designed to externalize internal physiological states, transforming invisible bodily data into perceptible feedback. Recent reviews on live biofeedback describe these systems as user interface elements that create a "physiological loop" between the user and the system [7, 12]. For example, "Living Bento" utilizes heart rate data to animate food, creating a shared dining experience driven by internal rhythms [1]. Similarly, D'Souza et al. explored augmenting co-located social play with biofeedback, finding that shared physiological signals can create moments of intense connection or playful competition [2]. Hornecker's frameworks on physicality in tangible interaction highlight how manipulating physical objects can support embodied cognition [4, 5]. Researchers have explored making intangible states such as emotions into tangibles through mixed reality and biometrics. For instance, "Tangible Intangibles" maps physiological signals to visual parameters in MR to support art therapy [9], and other works use "tangible body maps" to help users felt-sense their experiences [10].

We combine both the ideas of "making the internal external" and "tangible biofeedback" into the ideation of EERA, an interactive device that externalizes the user's eye as a manipulable and shareable object. EERA presents a real-time, magnified visualization of the user's eye, supports autonomous eye behaviors such as blinking, and enables the exchange of eye representations between participants.

Our design approach is informed by participatory design workshops across different organizations from different industries, such as technology, eye care pharmaceutical, and students from a school for the deaf and Hard-of-Hearing (DHH), for whom visual perception plays a particularly central role in communication and daily life. Through workshops conducted with these students and collaborating organizations, we explored alternative ways of thinking about future eye care beyond clinical or instructional models. This context emphasized the importance of making the eye itself visible, salient, and discussable, thus motivating a design that treats the eye not merely as a sensory organ, but as an object of perception and interaction.



Figure 2: Participatory design process. (a) A co-design workshop with DHH students, designers and researchers, exploring inclusive ways to raise awareness of eye health. (b) Documented ideation artifacts and discussion notes captured during the workshop, revealing recurring themes around enjoyment, accessibility, and embodied eye-care experiences.

We conducted preliminary user study, and collected 27 post-experience feedback responses, revealing key themes such as high user acceptance, embodied reflection, enhanced eye-care awareness, and increased social engagement through eye exchange. These findings suggest that EERA not only fosters a unique form of self-other comparison but also opens new avenues for promoting bodily awareness and conversations around eye health in an interactive, social context.

2 Design Process Of EERA

Participatory design with the DHH community has been shown to yield insights that benefit broader user groups by emphasizing clarity and visual expressiveness [6, 8]. We adopted a co-design approach, drawing on principles from the participatory design, in which researchers and designers with various background such as human ergonomics, technology and medical gathered together along with over 30 DHH high school students, for a 1-day workshop to brainstorm ideas as shown in Figure 2(a).

The workshop participants started with the problem statement "how to raise awareness about eye health inclusively?". The participants are separated into groups, each consisting of designer, researcher and 5 DHH students. The designer and researcher acted like a facilitator and mentor while the students explore their ideas without limitations to encourage creativity and innovation. Each brainstorming sessions lasted for 25 minutes and a 10 minute presentation was given after for every groups. Then finally, an overall discussion and presentation was conducted to conclude the ideation process.

All of the discussions and brainstorming are documented on Miro as shown in Figure 2(b). From there, the ideation workshop revealed several recurring sub-themes. Participants explored how eye care could be reframed as an **enjoyable and engaging experience** rather than a purely medical or corrective activity. At the same time, they raised critical questions about the balance between **novelty and practical usefulness**, noting that some ideas were interesting but might lack real-world value. **Accessibility and inclusivity** also emerged as important concerns, particularly whether proposed concepts would be meaningful or usable for people with visual impairments or older adults. Finally, many ideas expanded the notion of "seeing" beyond visual function to include emotional

awareness, self-reflection, and social perception, reflecting an early-stage exploration of eye health as an **embodied and experiential practice**.

From the extracted sub-themes, we defined four design goals: **fun and engaging**, **novel**, **inclusive**, and **embodied**. The Designers and we, the researchers collaboratively iterated on multiple prototypes to develop an intuitive interaction that accommodates diverse age groups and face shapes. The final design, shown in Figure 1(a), externalizes the eyes 1(b) as a playful 1(c-d) yet embodied object, supporting inclusive use while drawing attention to eye conditions through magnification.

3 System Implementation

Figure 3 shows EERA with 3 main parts: 1) The hairband which extends the camera to the front of the user's face; 2) The neck-hanging processing module that also serve as a holder for the 3) removable display module that displays the magnified user's eyes in real-time. For 1), the camera is positioned in front of the user's face, in between the user eyes not to occlude the views as well as distract the wearer during the interactions. The camera is able to capture a clear view of the user's eyes with an extended flexible gooseneck tube. The gooseneck tube allows minor adjustments for different face shapes and movements during use without losing the face track.

For 2), the processing module houses a Raspi 5, a battery and a wireless video transmitter and is worn hanging around the user neck. The video stream from the camera is fed into Raspi 5, which performs face landmarking with Mediapipe[3]. Landmark points from the left eyes to the right eyes horizontally and points from the eye brows to the nose bridge vertically are extracted and magnified to fill the entire display. Slight scaling and rotation compensations are also processed in real time to ensure full display coverage regardless of the eye sizes or improper camera placement. The video stream displaying the eyes is then transmitted wirelessly to the receiver found in the removable display module. In addition, we implemented a feature that detects the user's cue to enter the "blinking simulation" mode by closing their eyes for 3 seconds. During the "blinking simulation" mode, the real time video stream is paused and user's blinking eyes are displayed instead. Due to the limited computing capacity of the Raspi 5, we simulate the blinking of the eyes by alternating a series of photos for when the eyes are opened and when the eyes are closed which we captured during the usage. In 3), the removable display module displays the received video stream with negligible latency in 1080p resolution. The wireless video transmitter and receiver are paired exclusively by bluetooth connection. In this way, multiple EERA can work in the same space simultaneously without any crosstalk issues. Finally, we installed magnets around the edges of the display module as well as the processing module so that the display module can detach and attach to the processing module. This allows for different interactions such as freeing both users hand for games, attaching partner's "eyes" to one's self, or even attach the "eyes" on various magnetic surfaces such as white boards and refrigerators.

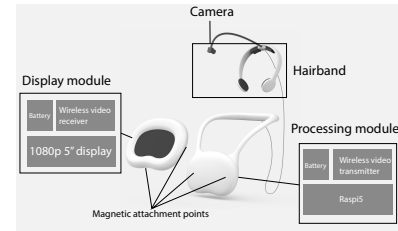


Figure 3: System overview of EERA. The system consists of three main components: a head-mounted camera attached via a flexible hairband, a neck-worn processing module containing a Raspberry Pi and wireless transmitter, and a detachable display module that presents a real-time, magnified visualization of the user's eyes.



Figure 4: Interactive experience flow. Participants are asked to perform a sequence of interactions: (1) participants observe their own externalized eye representation, (2) experience autonomous eye behaviors such as simulated blinking, (3) exchange eye representations with others, and (4) engage in an emotion guessing game designed to prompt self-other comparison and embodied reflection.

4 Preliminary User Study

We invited 27 participants (19 male, 8 female), aged 20 to 60 years ($M = 30$), to experience the device. Each session lasted approximately 5–10 minutes and was conducted in a table-top setting with a facilitator present. Participants wear EERA and interact using the detachable eye display through a sequence of playful activities. **(1) Self-observation.** Participants first hold and observe their own externalized eye representation. By moving and inspecting the display, they attend to eye movements and features that are typically unnoticed in everyday life. **(2) Autonomous blinking.** Participants activate a simulated blinking mode by closing their eyes for three seconds. This introduces a mild disruption of bodily agency, as the eye representation appears to behave independently from the participant's immediate control. **(3) Eye exchange.** Participants exchange eye representation with a partner and compare differences in appearance and behavior. The detachable, magnetic display supports holding or attaching a partner's eye, encouraging self-other comparison and social discussion. **(4) Social play.** Participants attach their eye representations on a whiteboard and each take turn to show an emotion based on the writings on the card they drew: happy, angry, sad and anxious. At the same time, the others guess what the correct emotions are. **Reflection.** The experience concludes with a brief, informal reflection prompted by the facilitator. Participants are invited to share any new awareness or perspectives on their eyes that emerged through the interaction.

We collected the post-experience questionnaires of EERA. The responses were collected on a 7-point Likert scale (1 = Strongly Disagree, 7 = Strongly Agree), along with an open-ended question for qualitative feedback. The questions which included in the appendix, are designed to assess baseline eye-care awareness, embodied reflection, social comparison, behavioral intention, and overall system acceptance, complemented by an open-ended reflection question. Quantitative results were analyzed using one-sample t-tests against the neutral midpoint (4.0).

User Acceptance. Participants reported high overall acceptance of the experience. The system was perceived as easy to understand ($M = 5.73$, $SD = 1.28$), and participants expressed strong willingness to recommend the experience to others ($M = 6.11$, $SD = 0.97$), indicating that EERA was engaging and approachable as an interactive system.

Embodied Reflection and Eye-Care Awareness. Participants reported significantly increased awareness of their own eyes after the experience ($M = 4.89$, $p < .01$). They also indicated that the experience led them to view their eyes as an object of care ($M = 5.04$, $p < .001$) and increased their motivation to care for their eyes in the future ($M = 5.15$, $p < .001$). Awareness increase was strongly correlated with viewing the eyes as an object of care ($r = .88$), suggesting that visual externalization played a key role in prompting reflection. The qualitative responses supported these findings. Participants frequently noted seeing their eyes in detail for the first time, which led to reflections on aging, fatigue, and everyday eye use (e.g., “Holding this ‘eye’ made me think about how much work it does for me every day”).

Social Engagement Through Eye Exchange. Exchanging eye representations heightened participants’ awareness of individual differences between their eyes and others’ ($M = 5.37$, $p < .001$) and moderately influenced how they perceived their interaction partner ($M = 4.63$, $p < .05$). Participants described the exchange as both playful and intimate, enabling close observation that would feel socially uncomfortable in direct eye contact (e.g., noticing blinking patterns or gaze stability).

5 Conclusion and Future Work

This work explored how externalizing a normally unseen body part (the eye) can support embodied reflection and social engagement around everyday eye care. Through the design and feedbacks of EERA, we showed how such device can create an embodied, shareable representation of the self. Findings from our user study suggest that physically holding and exchanging an externalized eye can increase awareness of eye conditions and prompt reflection on eye care, at the same time enable playful and intimate social interaction.

Future Work. First, longer-term deployments could examine whether repeated use leads to sustained changes in eye-care awareness or behavior beyond short-term reflection. Second, integrating richer sensing and actuation such as real-time blink rate analysis, gaze patterns, or tear-film indicators could further support meaningful reflection. Third, exploring use cases beyond facilitated tabletop settings, such as public exhibitions, classrooms, or clinical waiting areas, may reveal how externalized bodily representations function and benefit in everyday social environments. Finally, expanding

participatory design with additional communities, including people with visual impairments or chronic eye conditions, could inform more inclusive and context-sensitive designs. Together, this work points to the potential of a tangible, embodied biofeedback systems as a design space for supporting bodily awareness.

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